

Minimisation problem

The constraint is $x_1 + a_2 x_2 = 1$ with $x_1, x_2 \geq 0$, and we want to minimize $T = x_1 + x_2$. By linear-programming theory the optimal solution lies at a vertex of the feasible region ¹. Here the two relevant corner points are $(x_1, x_2) = (1, 0)$ giving $T = 1$, and $(0, 1/a_2)$ giving $T = 1/a_2$. Since the given code makes a_2 slightly greater than 1 (it equals $1 + 2 \cdot 72^{-10^{24}} + \dots \approx 1$), we have $1/a_2 < 1$. Thus the minimiser is $(x_1, x_2) \approx (0, 1)$.

The proposed point $(0.1, 0.9)$ differs from $(0, 1/a_2) \approx (0, 1)$ by $|0.1 - 0| = 0.1$ in the first coordinate and $|0.9 - 1| \approx 0.1$ in the second. Its infinity-norm distance is $\max(0.1, 0.1) = 0.1$, which is well below 0.15.

Answer: Yes - $(0.1, 0.9)$ is about $0.1 < 0.15$ away (in the ℓ_∞ norm) from the true minimiser $(0, 1/a_2)$

¹ .

¹ optimization - Why does the maximum/minimum of linear programming occurs at a vertex? - Mathematics Stack Exchange

<https://math.stackexchange.com/questions/896388/why-does-the-maximum-minimum-of-linear-programming-occurs-at-a-vertex>